RepIt – Machine Learning Technical Overview

(Confidential – Shared Under NDA)

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1. System Overview

RepIt is a dual-device motion analysis system:

Device 1: Worn on the abdomen (under the chest)

Device 2: Mounted on exercise equipment

Both devices collect IMU-based motion data (accelerometer + gyroscope).

Data streams are time-synchronized and transmitted via BLE to a mobile application.

The ML system processes correlated dual-stream motion data to provide:

- Activity classification
- Form correction feedback
- RPE (fatigue) estimation
- Injury-risk detection
- Long-term adaptive personalization

2. Current Hardware Context

Current prototype:

- ESP32-C3
- MPU6050 / BMI160 IMU
- Basic BLE data transmission working
- No structured ML pipeline yet

Inference will initially run on mobile.

3. Phase 1 (MVP) ML Objectives

Core Goals

• Motion classification

• Dual-stream pattern correlation

• Form deviation detection

• Basic RPE estimation

• Real-time mobile inference

4. Algorithmic Requirements

A. Signal Processing Layer

• Noise filtering (low-pass / high-pass)

• Sensor fusion (accelerometer + gyroscope)

• Drift compensation

• Sliding window segmentation

• Dual-stream synchronization validation

B.Feature Engineering

• Time-domain features:

- Mean, variance, RMS
- Zero-crossings
- Peak detection
- Signal magnitude area

• Frequency-domain features:

- FFT-based spectral energy
- Dominant frequencies
- Harmonic behavior

• Cross-device features:

- Cross-correlation
- Phase alignment
- Relative motion patterns
- Stability & symmetry metrics

C. Modeling Architecture

We are open to architectural recommendations, but considering:

• CNN (local pattern extraction)

• LSTM / GRU (temporal modeling)

• Hybrid CNN-LSTM

• Dual-input fusion networks

• Lightweight neural models optimized for mobile

• Possibly tree-based baselines for comparison

D. Form Correction

• Baseline movement modeling

• Deviation scoring

• Regression-based form quality metric

• Probabilistic error detection

E. RPE Estimation

• Fatigue modeling over time

• Velocity degradation tracking

• Stability variation tracking

• Regression-based intensity estimation

F. Injury Prevention (Phase 2 Direction)

• Anomaly detection

• Asymmetry detection

• Overload pattern recognition

• Longitudinal instability tracking

G. Adaptive Learning (Future Direction)

• User-specific baseline calibration

• Incremental learning

• Model fine-tuning per user

• Personalization layer

5. Constraints

• Dual synchronized time-series

• Moderate sampling frequency (TBD)

- Mobile-side inference initially
- Full-day hardware battery constraint
- Total development window: 6 weeks

We are senior university students building RepIt as an early-stage startup.

We strongly believe in this idea and are committed to building it professionally and properly. Our budget is limited at this stage, but we are investing carefully to create a real scalable product.

We are looking for someone who can think architecturally, move efficiently, and potentially grow with the project long-term.